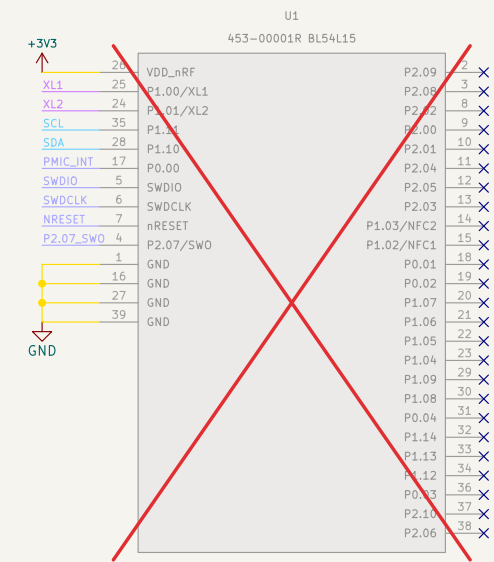


RADIO / MCU – BL54L15

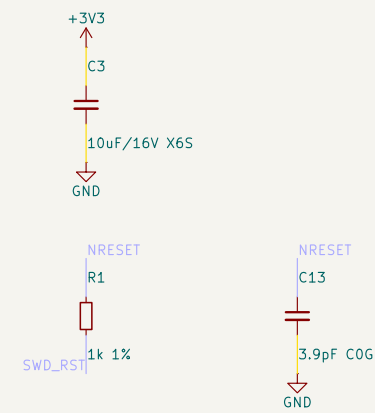


453-00001R: integrated PCB antenna.

U1 DNP is for JLC assembly only; hand fit for use.

SDA / SCL: configure GPIO selection in firmware.

MODULE SUPPLY / FILTER

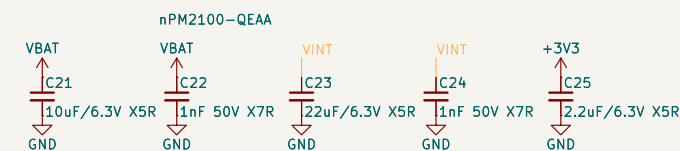
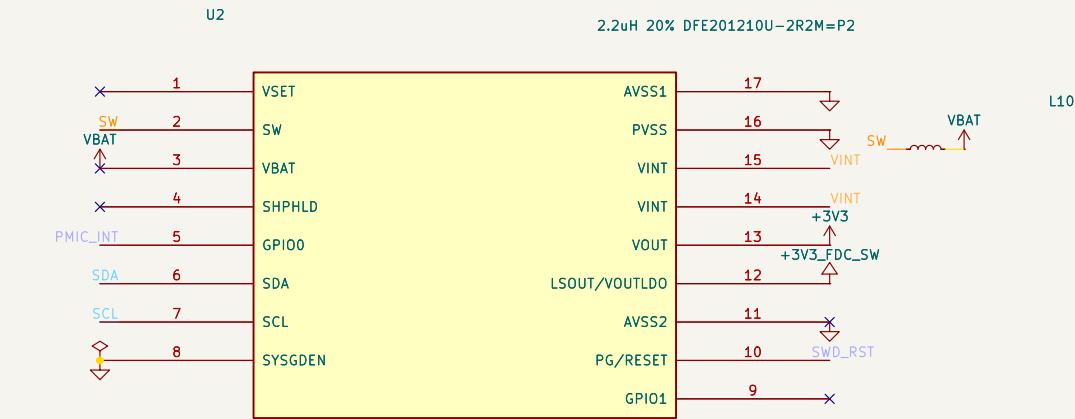


C3: external module bulk capacitor.

R1 / C13: module reset input network.

Keep antenna copper / component keepout clear.

POWER – nPM2100

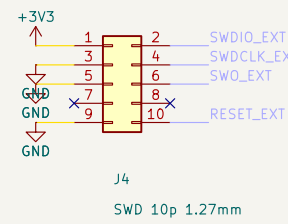


VBAT -> boost -> +3V3

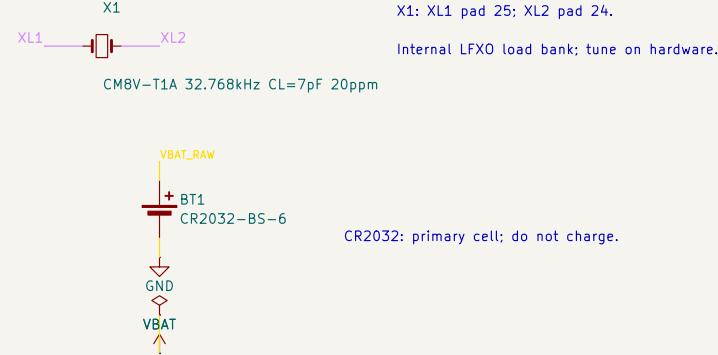
+3V3_FDC_SW supplies FDC1004 only.

GPI00: PMIC_INT | GPI01: unused (NC).

SWD PROGRAMMING



32 kHz / BATTERY



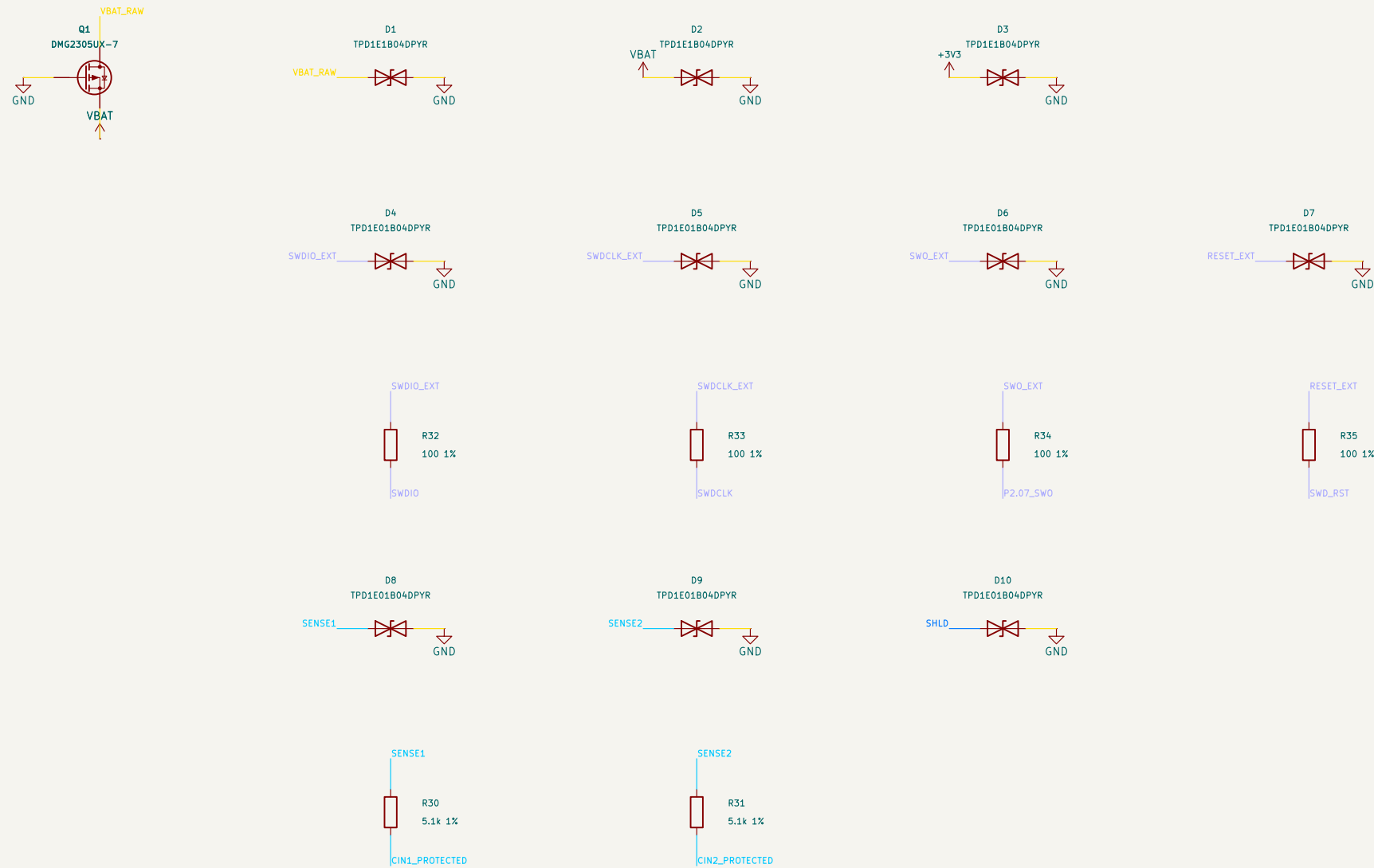
FIDUCIALS

-  FID1
Fiducial
-  FID2
Fiducial
-  FID3
Fiducial

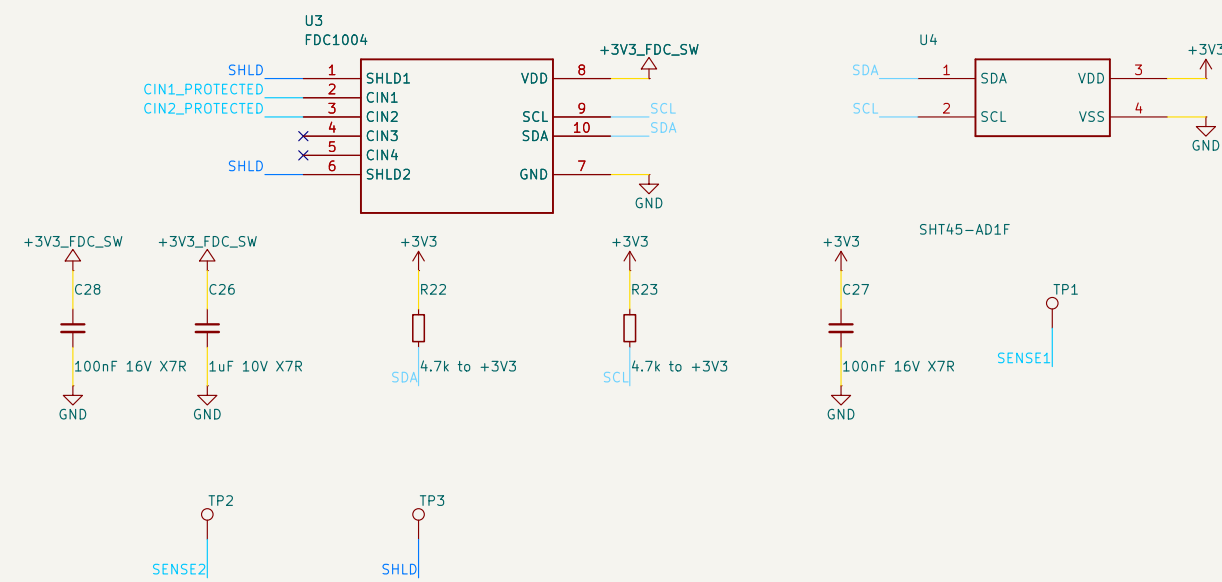
INPUT PROTECTION

PMOS reverse polarity; ESD clamps + series resistors.

No sustained overvoltage cutoff or battery fuse



MOISTURE + TEMPERATURE / HUMIDITY



FDC1004: switched supply. SHT45: main +3V3 supply.

R22 / R23: shared I2C pull-ups to +3V3.

CIN1 / CIN2 via 5.1k protection; SHLD driven by FDC.

Hardware rationale: [HARDWARE.md](#) | Electrical review: [docs/design-review-2026-09-05/U1](#) is hand fitted after JLC assembly; firmware must select the labeled GPIOs
VBAT: protected battery | +3V3: main rail | +3V3_FDC_SW: switched sensing rail
BL54L15 + nPM2100 + FDC1004 + SHT45 | CR2032 | BTHome v2

Sheet: /
File: nrf-moisture-sensor.kicad_sch

Title: nRF Moisture Sensor

Size: A2	Date: 2026-09-05	Rev: C
KiCad E.D.A. 10.99.0-3397-g20ddeb141f		Id: 1/1